

# Data Science & GenAI Assessment Report

Store Sales Analysis | Policy Q&A Bot

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**Tool:** Google Colab (Python 3.10)

**Approach:** Random Forest | TF-IDF + Cosine Similarity

## Assignment 1 — Store Sales Analysis & Prediction

The goal was to analyze one full year of daily store sales data, engineer meaningful features, train a forecasting model, evaluate it on the last 30 days, predict the next 7 days, and extract business insights.

### Dataset

**sales\_data.csv** — 366 rows (full year 2024). Four columns: Date, Sales (daily revenue in Rs.), Promotion (1 if promotion ran, else 0), DayOfWeek (1=Mon to 7=Sun). **No missing values.** Promotion ran on 71 days (19.4%).

### Key EDA Findings

| Metric                 | Value               | Finding                                |
|------------------------|---------------------|----------------------------------------|
| No-Promotion Avg Sales | Rs. 200.38          | Baseline daily revenue                 |
| Promotion Avg Sales    | Rs. 248.27          | +23.9% lift on promo days              |
| Best Day of Week       | Wednesday           | Rs. 227.66 avg                         |
| Worst Day of Week      | Saturday            | Rs. 185.00 avg — weekends underperform |
| Sales Range            | Rs. 149.93 – 286.29 | No extreme outliers                    |

Promotions deliver a consistent **+23.9% revenue lift**. Despite running on only 1 in 5 days, they drive disproportionate revenue. Weekends (Sat & Sun) average Rs. 189.82 vs. a weekday average of Rs. 217.54 — a Rs. 27.72 gap — suggesting a weekday-centric customer base.

### Feature Engineering

Nine features were created from the raw data. Lag and rolling features use **.shift(1)** before computation to prevent data leakage — the model never uses today's sales to predict today.

| Feature                                          | Description                                                   |
|--------------------------------------------------|---------------------------------------------------------------|
| lag_1                                            | Previous day's sales — strongest single predictor             |
| rolling_avg_7                                    | 7-day rolling average (weekly trend)                          |
| rolling_avg_14                                   | 14-day rolling average (fortnightly trend)                    |
| Promotion                                        | Binary promotion flag from raw data                           |
| DayOfWeek, Month, WeekOfYear, Quarter, IsWeekend | Calendar decomposition — encodes seasonal & cyclical patterns |

### Model — Random Forest Regressor

An ensemble of **200 decision trees**. Chosen because it handles non-linear relationships, needs no feature scaling, and provides built-in feature importance. Split was strictly chronological: **train = Jan 15 – Dec 1 (322 days), test = Dec 2 – Dec 31 (last 30 days)**. No random split — that would leak future data into training.

### Evaluation Results

|                                         |                                        |                                  |                                 |
|-----------------------------------------|----------------------------------------|----------------------------------|---------------------------------|
| <b>Rs. 12.67</b><br>Mean Absolute Error | <b>Rs. 16.49</b><br>Root Mean Sq Error | <b>6.54%</b><br>Mean Abs % Error | <b>6.20%</b><br>MAE / Avg Sales |
|-----------------------------------------|----------------------------------------|----------------------------------|---------------------------------|

A MAPE of **6.54%** is well below the 10% benchmark for retail forecasting. The RMSE-to-MAE ratio of 1.30x confirms no large systematic errors — residuals scatter randomly around zero. **Promotion** was the most important feature (50.2% importance), followed by **DayOfWeek** (24.0%), confirming EDA findings.

### 7-Day Sales Forecast (Jan 1–7, 2025)

| Date       | Day       | Predicted Sales (Rs.) |
|------------|-----------|-----------------------|
| 2025-01-01 | Wednesday | 218.19                |
| 2025-01-02 | Thursday  | 210.83                |
| 2025-01-03 | Friday    | 199.56                |
| 2025-01-04 | Saturday  | 178.90                |
| 2025-01-05 | Sunday    | 186.24                |
| 2025-01-06 | Monday    | 206.99                |
| 2025-01-07 | Tuesday   | 218.69                |

7-Day Total: Rs. 1,419.40 | Daily Average: Rs. 202.77 | Uncertainty: +/- Rs. 12.67 (one MAE)

5 Business Insights

| #  | Insight & Action                                                                                                                                                                    |
|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. | Promotions are the #1 revenue lever (+23.9% lift). Increasing frequency during slow weeks can unlock significant revenue without new customers.                                     |
| 2. | Wednesday is peak (Rs. 227.66), Saturday is worst (Rs. 185.00). Maximise staff and stock mid-week; run weekend deals to close the Rs. 42.66 gap.                                    |
| 3. | Weekends consistently underperform weekdays — unusual for retail. Investigate store hours, demographics, or competition and introduce weekend loyalty offers.                       |
| 4. | Promotion feature holds 50.2% of model importance — the single biggest lever. A live daily dashboard to trigger same-day promotions on slow days is high-ROI.                       |
| 5. | Sales follow a repeatable seasonal cycle. Stock up 3-4 weeks before peak months and run clearance in troughs. The 6.54% MAPE makes this model reliable for monthly demand planning. |

## Assignment 2 — Simple Policy Q&A; Bot

Build a Q&A; system that retrieves answers from three company policy documents using classical NLP — no paid APIs, no GPU models, no internet required.

### Documents

| File              | Sentences | Key Topics                                                                                    |
|-------------------|-----------|-----------------------------------------------------------------------------------------------|
| leave_policy.txt  | 6         | Paid leave (15 days), Sick (10 days), Maternity (26 wk), Casual (3 days max), carry-forward   |
| it_policy.txt     | 5         | VPN, Passwords, Company laptop, Antivirus, USB devices                                        |
| travel_policy.txt | 6         | Economy flights, Hotel Rs. 5000/night, Local bills, International approval, Personal expenses |

Each file was split into individual sentences by line, stripping numbering and blank headers. This produced **17 clean sentences** — the atomic units the system searches over.

### Approach — Two-Stage Retrieval

A single-stage TF-IDF approach failed on ambiguous queries. For example, "sick leaves **allowed**" scored higher on the IT Policy sentence "USB devices are not **allowed**..." than on the correct Leave Policy answer, because the word "allowed" is rare and gets a high TF-IDF weight. The fix: restrict the search space before running TF-IDF.

| Stage   | Method                     | What it does                                                                                                                                           |
|---------|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| Stage 1 | Keyword Domain Detection   | Count domain-specific keywords (e.g. "sick", "leave", "maternity") in the question. Route to the matching policy. Zero matches = fallback immediately. |
| Stage 2 | TF-IDF + Cosine Similarity | Convert question to TF-IDF vector. Compare against sentences from the detected domain only. Return best match if score > 0.15, else fallback.          |

**TF-IDF Config:** stop\_words="english" | ngram\_range=(1,2) | Matrix: 17 sentences x 130 unique terms | Similarity threshold: 0.15

### Test Results — 11 / 11 Passed

| Question                                | Domain | Answer                                 | Score | Status |
|-----------------------------------------|--------|----------------------------------------|-------|--------|
| How many paid leaves per year?          | Leave  | Entitled to 15 paid leaves per year    | 0.616 | PASS   |
| What is maternity leave duration?       | Leave  | Maternity leave is 26 weeks            | 0.621 | PASS   |
| Can I carry forward unused leaves?      | Leave  | Cannot be carried forward to next year | 0.513 | PASS   |
| Is VPN required for remote work?        | IT     | VPN is mandatory for remote access     | 0.338 | PASS   |
| Can I use a USB drive at work?          | IT     | USB devices not allowed without IT...  | 0.258 | PASS   |
| What device for official work?          | IT     | Company-issued laptop must be used     | 0.507 | PASS   |
| Hotel reimbursement limit per night?    | Travel | Limit is Rs. 5000 per night            | 0.739 | PASS   |
| What class for flight bookings?         | Travel | Must be economy class                  | 0.378 | PASS   |
| Approval needed for international trip? | Travel | Requires prior managerial approval     | 0.564 | PASS   |
| Sick leaves allowed                     | Leave  | Sick leave entitlement is 10 days      | 0.191 | PASS   |

| Question                        | Domain | Answer                       | Score | Status |
|---------------------------------|--------|------------------------------|-------|--------|
| What is the canteen menu today? | None   | Information not available... | N/A   | PASS   |

The system correctly answered all 9 policy questions and blocked both out-of-scope queries. The previously failing "sick leaves allowed" query now returns the correct Leave Policy answer (score 0.191) because Stage 1 routes it to Leave Policy before TF-IDF runs, eliminating the IT Policy interference.

## Conclusion

| Assignment      | Method                                         | Key Result                                                   |
|-----------------|------------------------------------------------|--------------------------------------------------------------|
| 1 — Store Sales | Random Forest (200 trees, chronological split) | MAE Rs.12.67   MAPE 6.54%   7-day forecast Rs.1,419.40       |
| 2 — Policy Q&A; | Two-stage TF-IDF + Cosine Similarity           | 11/11 test cases passed   Fully CPU-based   <1 sec per query |

Both solutions run entirely on free Google Colab with no paid APIs. Code is modular, reproducible, and well-documented. The Promotion flag emerged as the dominant signal in Assignment 1 (50.2% importance), and domain-restricted retrieval was the critical fix in Assignment 2. Together, these demonstrate practical application of machine learning and classical NLP to real business problems.

Tools: Python 3.10 · pandas · numpy · matplotlib · seaborn · scikit-learn (RandomForestRegressor, TfidfVectorizer, cosine\_similarity) · Google Colab